Box Plots

Name _____ Date _____ No calculator Version 1

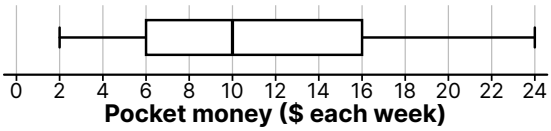
Scan to practise this topic online

KEY WORDS box plot *(kauwhata pouaka)* · median *(tau waenga)* · lower quartile *(hauwhā raro)*

A Reading a box plot ●○○

Remember: The box runs from the lower quartile to the upper quartile, with a line at the median. The whiskers reach the minimum and the maximum. The interquartile range (IQR) is the width of the box.

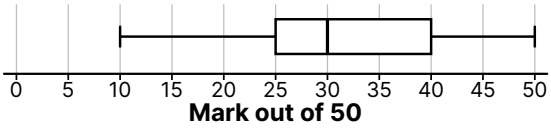
1. The box plot shows the pocket money, in dollars, that a group of students get each week.



a) Minimum _____ **b)** Lower quartile ____ **c)** Median _____ **d)** Upper quartile ____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

2. The box plot shows the marks of a class in a spelling test out of 50.



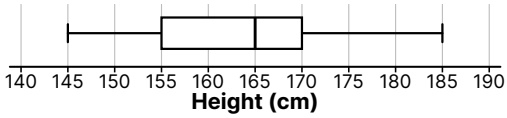
a) Minimum _____ **b)** Lower quartile ____ **c)** Median _____ **d)** Upper quartile ____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

B The quarters of a box plot ●●○

Remember: The five numbers cut the data into four parts. Each part holds a quarter (25%) of the values, even when the parts are different widths.

3. The box plot shows the heights of the students in a class.



a) What percentage of the students are less than 165 cm?

b) What percentage are more than 170 cm? _____

c) What percentage are between 155 cm and 170 cm? ____

d) What percentage are less than 155 cm? _____

e) What percentage are more than 155 cm? _____

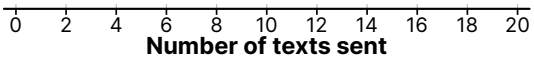
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C Drawing a box plot ●●○

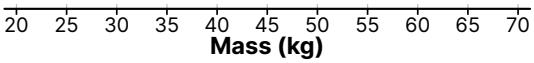
Remember: Mark the five numbers above the axis. Draw the box from LQ to UQ, a line at the median, and whiskers out to the minimum and maximum.

4. Draw a box plot above each axis.

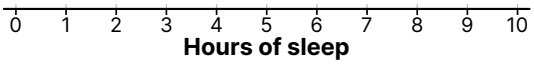
a) Minimum 2, LQ 6, median 10, UQ 14, maximum 18



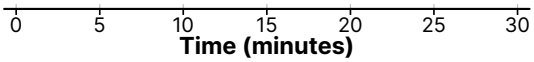
b) Minimum 30, LQ 40, median 45, UQ 55, maximum 65



c) Minimum 5, LQ 7, median 8, UQ 9, maximum 10



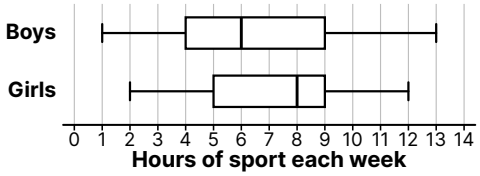
d) Data: 5, 10, 10, 15, 20, 25, 30. Find the five-point summary first.



D Comparing two box plots ●●●

Remember: Compare the middles (the medians) and the spreads (the ranges and interquartile ranges).

5. The box plots show the hours of sport that a group of boys and a group of girls play each week.



a) Which group has the higher median? Write down both medians.

b) Find the range of each group.

c) Find the interquartile range of each group.

d) In which group is the middle half of the data more spread out?

e) True or false: more than half of the Girls group play more than 9 hours.

f) Describe the shape of each group.

★ Extension

6. A box plot has a lower quartile of 6 and an upper quartile of 15. Its minimum is 2 and its range is 30.

a) Find the interquartile range.

b) Find the maximum.

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Box Plots

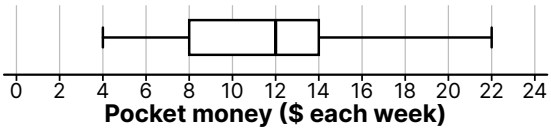
Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS box plot *(kauwhata pouaka)* · median *(tau waenga)* · lower quartile *(hauwhā raro)*

A Reading a box plot ●○○

Remember: The box runs from the lower quartile to the upper quartile, with a line at the median. The whiskers reach the minimum and the maximum. The interquartile range (IQR) is the width of the box.

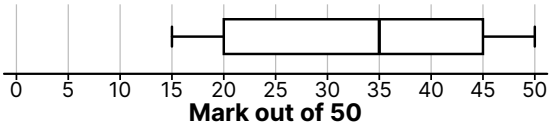
1. The box plot shows the pocket money, in dollars, that a group of students get each week.



a) Minimum _____ **b)** Lower quartile ____ **c)** Median _____ **d)** Upper quartile ____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

2. The box plot shows the marks of a class in a spelling test out of 50.



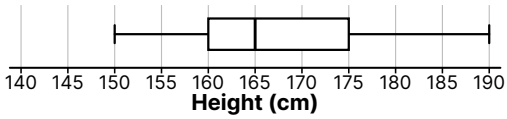
a) Minimum _____ **b)** Lower quartile ____ **c)** Median _____ **d)** Upper quartile ____

e) Maximum _____ **f)** Range _____ **g)** IQR _____

B The quarters of a box plot ●●○

Remember: The five numbers cut the data into four parts. Each part holds a quarter (25%) of the values, even when the parts are different widths.

3. The box plot shows the heights of the students in a class.



a) What percentage of the students are less than 165 cm?

b) What percentage are more than 175 cm? _____

c) What percentage are between 160 cm and 175 cm? ____

d) What percentage are less than 160 cm? _____

e) What percentage are more than 160 cm? _____

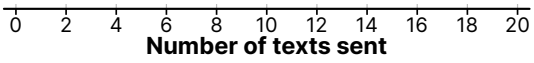
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C Drawing a box plot ●●○

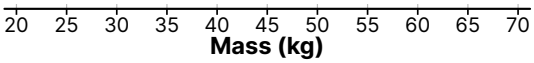
Remember: Mark the five numbers above the axis. Draw the box from LQ to UQ, a line at the median, and whiskers out to the minimum and maximum.

4. Draw a box plot above each axis.

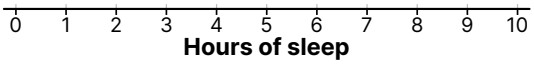
a) Minimum 4, LQ 8, median 10, UQ 16, maximum 20



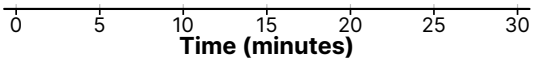
b) Minimum 25, LQ 35, median 50, UQ 55, maximum 70



c) Minimum 4, LQ 6, median 7, UQ 9, maximum 10



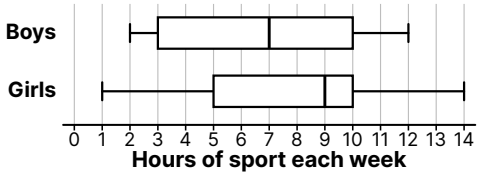
d) Data: 5, 5, 10, 15, 25, 25, 30. Find the five-point summary first.



D Comparing two box plots ●●●

Remember: Compare the middles (the medians) and the spreads (the ranges and interquartile ranges).

5. The box plots show the hours of sport that a group of boys and a group of girls play each week.



a) Which group has the higher median? Write down both medians.

b) Find the range of each group.

c) Find the interquartile range of each group.

d) In which group is the middle half of the data more spread out?

e) True or false: more than half of the Girls group play more than 10 hours.

f) Describe the shape of each group.

★ Extension

6. A box plot has a lower quartile of 8 and an upper quartile of 20. Its minimum is 4 and its range is 32.

a) Find the interquartile range.

b) Find the maximum.

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